

Title of Invention:

AI-DRIVEN IIoT FRAMEWORK FOR REAL-TIME INDUSTRIAL EQUIPMENT MONITORING

Drawings:

FIG. 1: Overall architecture of the AI-driven IIoT framework.

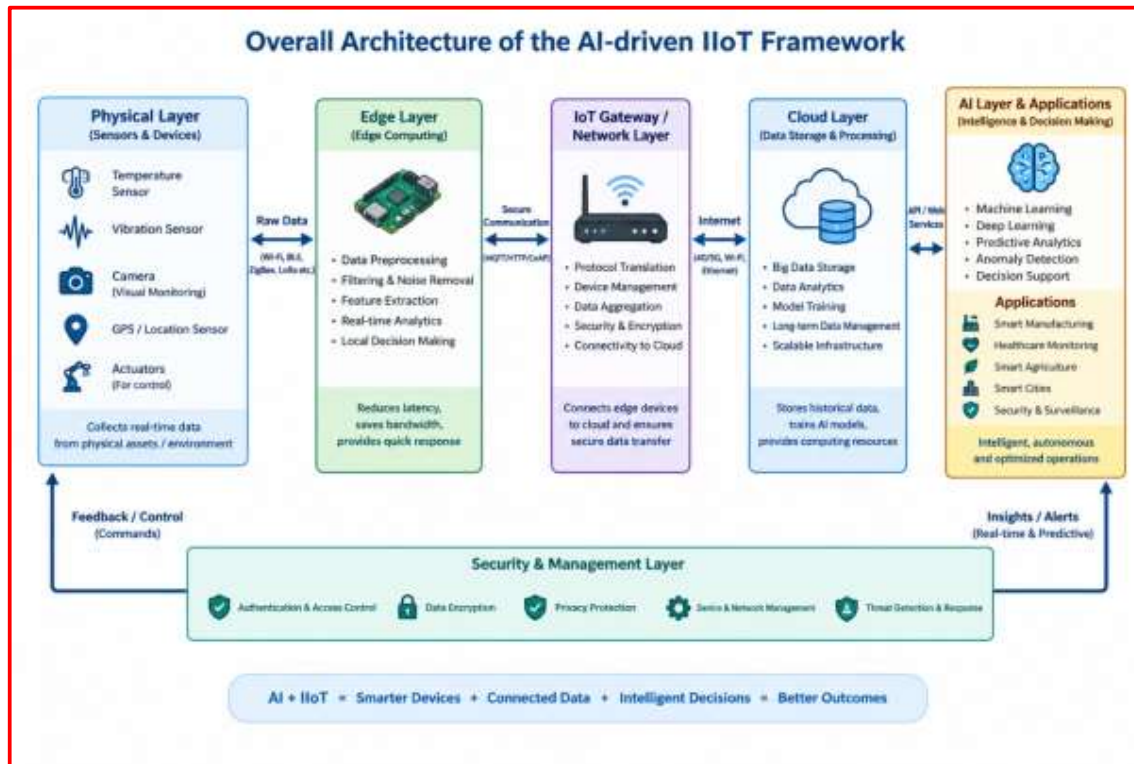


FIG. 2: Industrial sensor and data-acquisition subsystem.

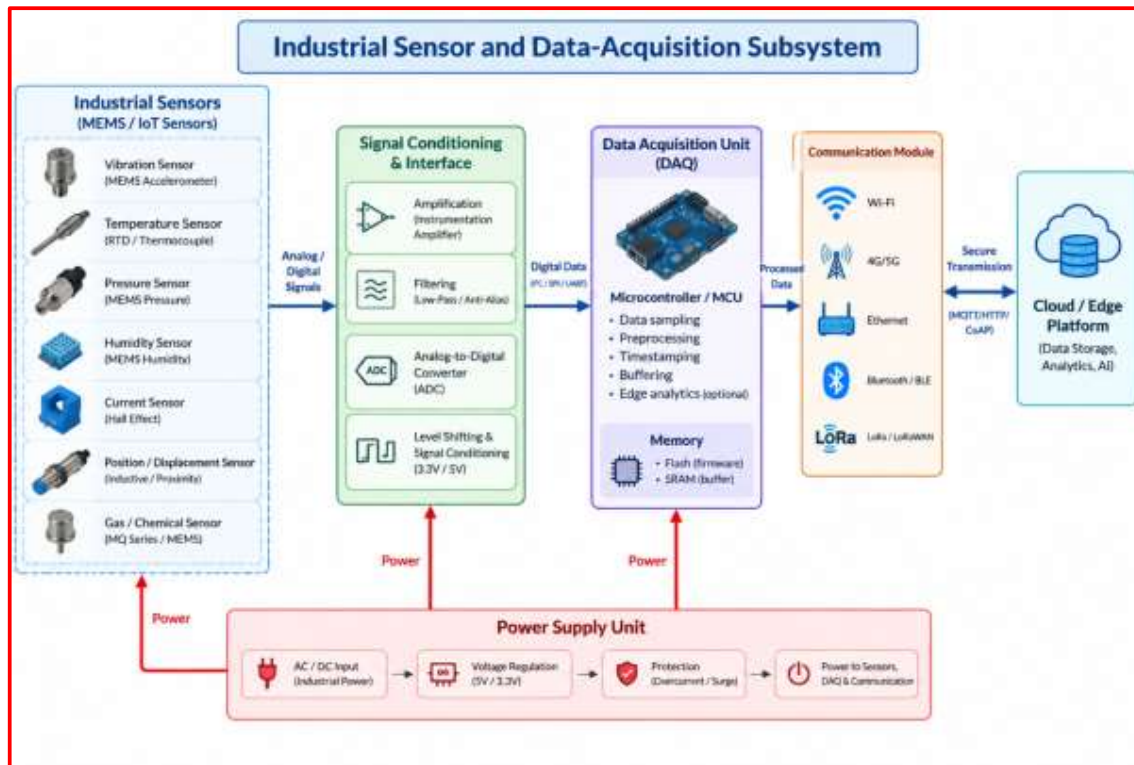


FIG. 3: Architecture of the IIoT edge gateway.

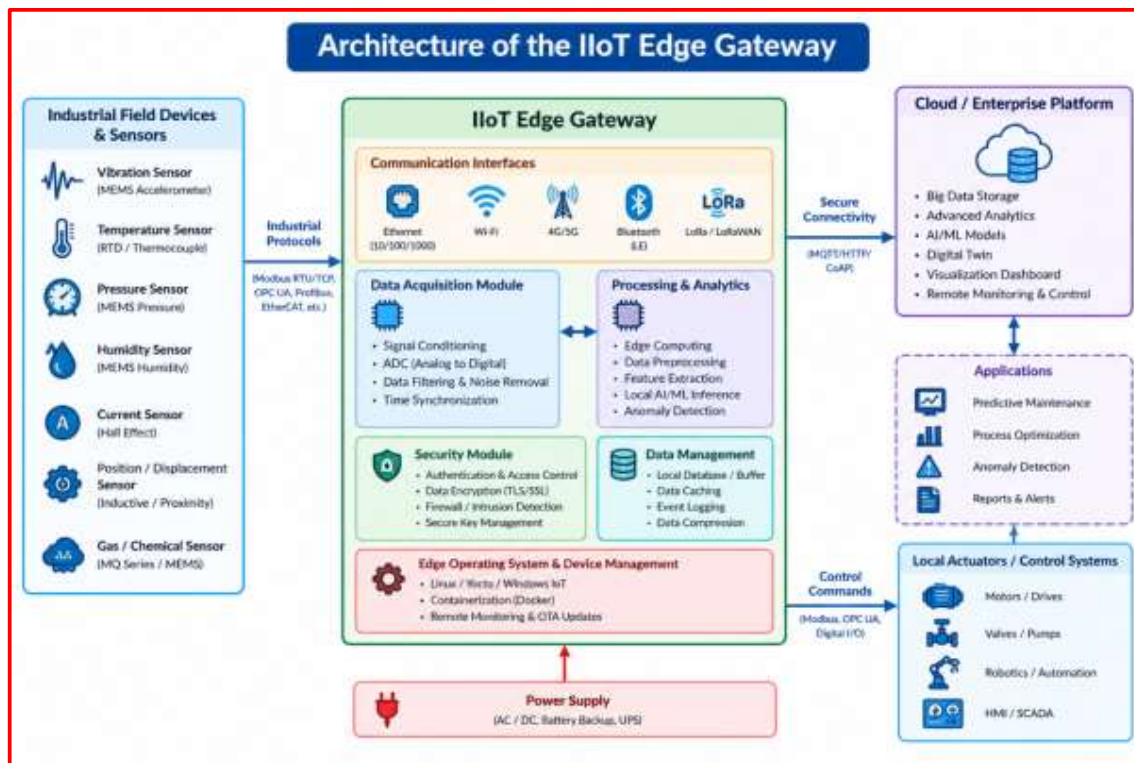


FIG. 4: AI/ML-based anomaly detection and equipment-health assessment process.

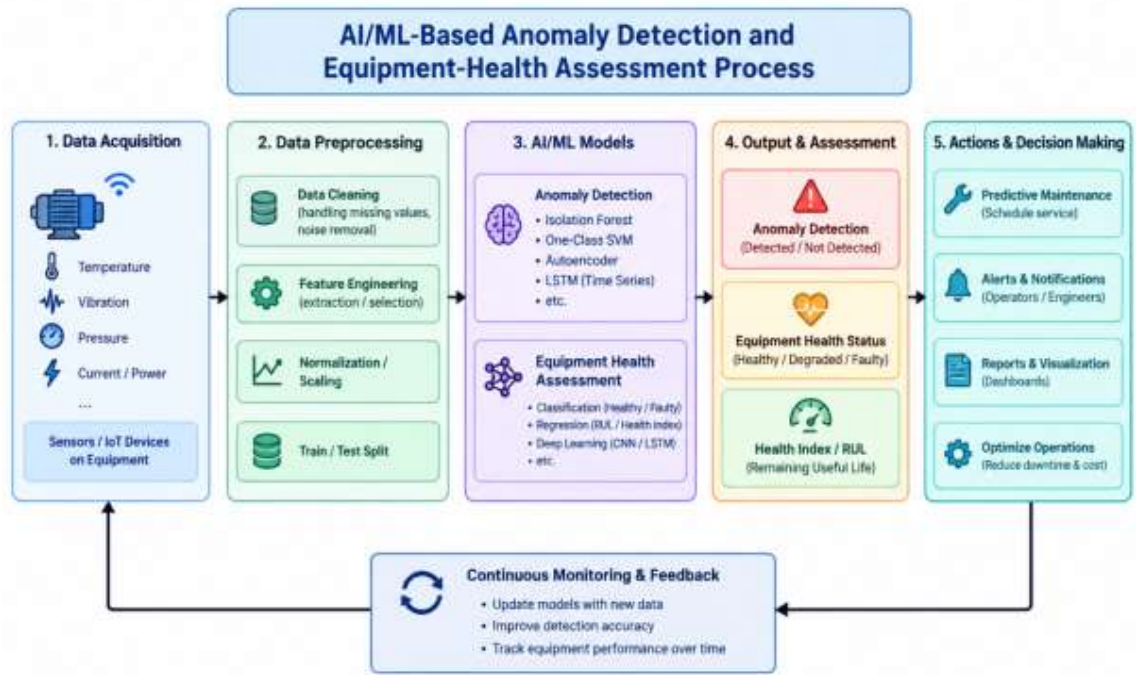


FIG. 5: Predictive maintenance and fault-prediction workflow.

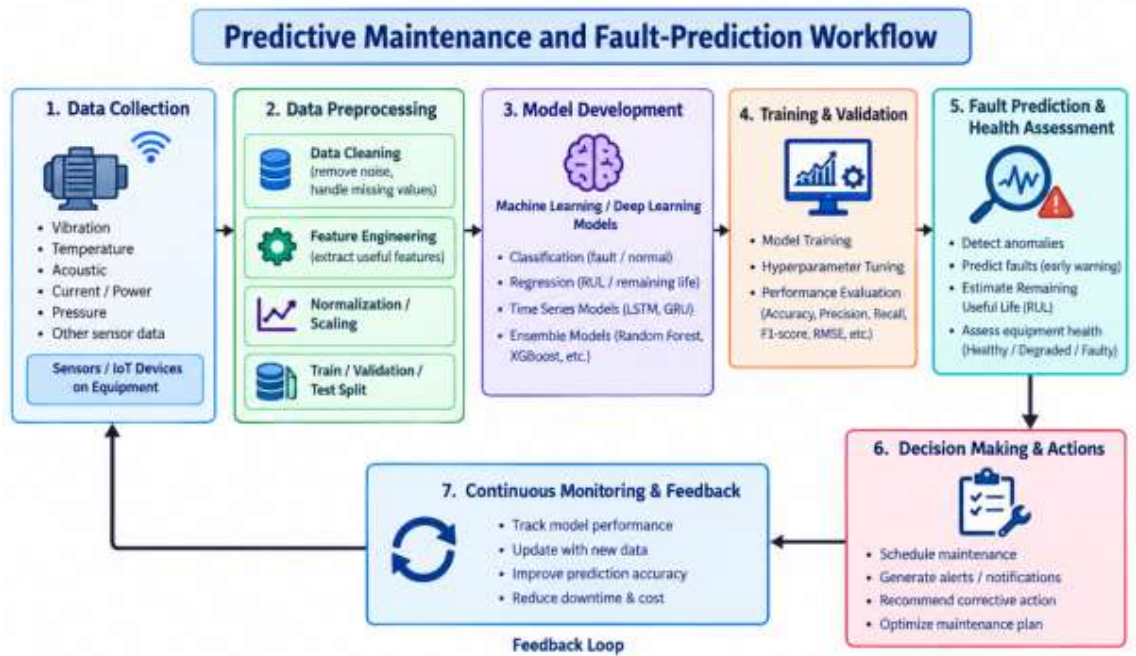
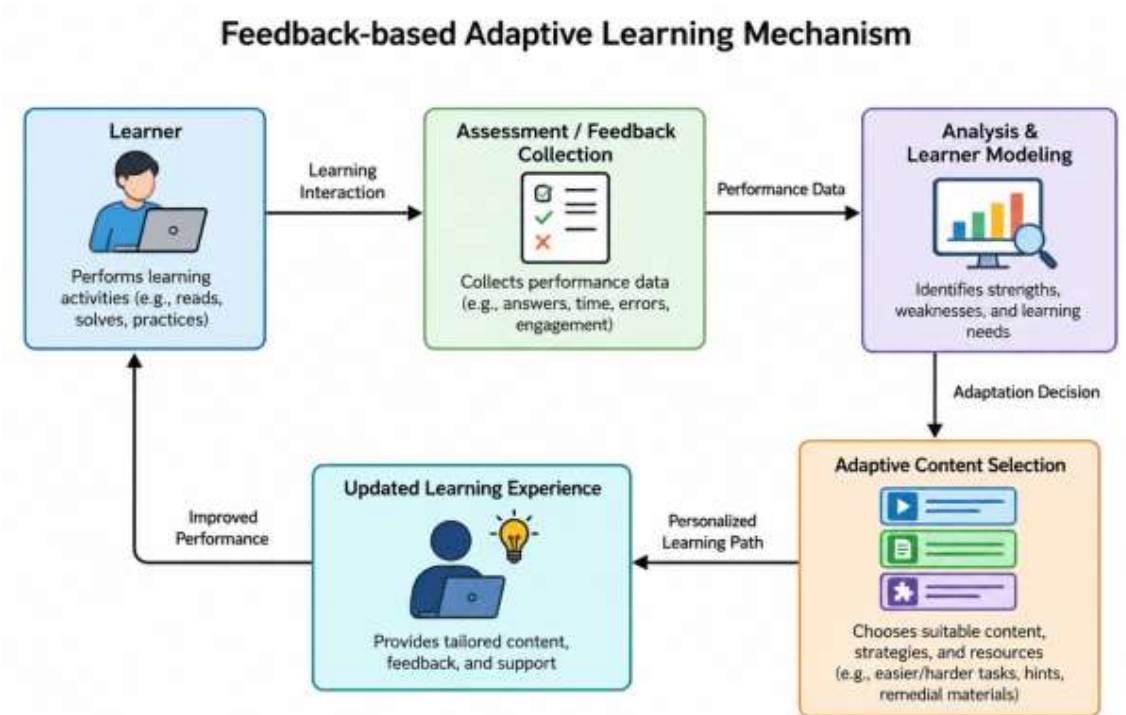


FIG. 6: Industrial dashboard showing equipment health, anomaly score and alerts.



FIG. 7: Feedback-based adaptive learning mechanism.



Explanation of Figures

FIG. 1 – Overall Architecture of the AI-Driven IIoT Framework

1. FIG. 1 illustrates the overall architecture of the proposed AI-driven Industrial Internet of Things (IIoT) framework for real-time industrial equipment monitoring.
2. The architecture consists of the industrial equipment layer, sensor and data-acquisition layer, IIoT edge gateway, communication layer, AI/ML analytics layer, data-storage layer, and visualization and decision-support layer.
3. Sensors acquire parameters such as vibration, temperature, pressure, current, voltage, acoustic signals, and rotational speed from industrial equipment.
4. The edge gateway preprocesses the acquired data and transmits relevant information to the AI/ML analytics engine through a communication module.
5. The AI/ML engine analyses the processed data to assess equipment health, detect abnormalities, predict faults, and provide maintenance-related information through the decision-support module.

FIG. 2 – Industrial Sensor and Data-Acquisition Subsystem

1. FIG. 2 illustrates the sensor and data-acquisition subsystem used for collecting real-time operating parameters from industrial equipment.
2. A plurality of sensors may include vibration sensors, MEMS accelerometers, temperature sensors, pressure sensors, current and voltage sensors, acoustic sensors, and rotational-speed sensors.
3. The sensors continuously or periodically acquire equipment operating data according to the requirements of the particular industrial asset.
4. The acquired data may include vibration, temperature, pressure, electrical, acoustic, and rotational-speed information represented as a multi-parameter data set.
5. The subsystem provides the collected sensor information to the IIoT edge gateway for filtering, normalization, feature extraction, anomaly detection, and subsequent AI/ML analysis.

FIG. 3 – Architecture of the IIoT Edge Gateway

1. FIG. 3 illustrates the architecture of the IIoT edge gateway responsible for acquiring and locally processing sensor data.
2. The gateway comprises processing capability, memory, communication interfaces, and data-processing functionality for handling incoming sensor information.
3. The gateway performs operations including timestamp synchronization, signal filtering, noise removal, missing-data identification, normalization, and feature extraction.
4. Local anomaly detection and data compression may also be performed at the gateway before transmission to the remote analytics platform.
5. Edge processing reduces unnecessary transmission of raw sensor data and enables faster responses to critical equipment conditions.

FIG. 4 – AI/ML-Based Anomaly Detection and Equipment-Health Assessment Process

1. FIG. 4 illustrates the AI/ML-based process for detecting abnormal equipment behaviour and assessing the health condition of industrial equipment.
2. Processed sensor signals are converted into statistical, temporal, and frequency-domain features forming a feature vector.
3. The feature vector is supplied to an AI/ML analytics engine, which may employ classification, regression, neural-network, tree-based, clustering, or other machine-learning models.
4. The system generates an anomaly score by comparing the current feature profile with a learned baseline and calculates an equipment health index using sensor features, model outputs, and historical information.
5. The equipment condition can be classified into normal, warning, abnormal, or critical states to support timely identification of degradation and abnormal operation.

FIG. 5 – Predictive Maintenance and Fault-Prediction Workflow

1. FIG. 5 illustrates the predictive maintenance and fault-prediction workflow of the proposed framework.
2. The system analyses temporal sensor characteristics and historical equipment information to estimate the likelihood of future equipment failure.
3. Potential faults may include bearing degradation, motor overheating, excessive vibration, shaft misalignment, electrical overload, lubrication abnormalities, pressure abnormalities, and imbalance.
4. The prediction module may generate a fault probability or predicted degradation level for the monitored equipment.
5. Based on the predicted fault and its severity, the decision-support module generates maintenance alerts and recommendations, thereby supporting predictive rather than solely reactive maintenance.

FIG. 6 – Industrial Dashboard Showing Equipment Health, Anomaly Score and Alerts

1. FIG. 6 illustrates the industrial dashboard used to present equipment-health information, anomaly scores, alerts, and related analytics.
2. The dashboard can display the real-time equipment status, equipment health index, anomaly score, fault category, and severity level.
3. It can provide maintenance alerts and recommendations to operators and maintenance personnel.
4. Historical trends can be displayed to assist in analysing equipment behaviour and supporting maintenance decisions.
5. The dashboard may be accessed through an industrial workstation, computer, tablet, or mobile interface, enabling practical monitoring and decision support.

FIG. 7 – Feedback-Based Adaptive Learning Mechanism

1. FIG. 7 illustrates the feedback-based adaptive learning mechanism through which actual equipment conditions and maintenance outcomes are fed back to the AI/ML analytics engine.
2. The system records subsequent equipment conditions, inspection results, maintenance events, and actual failure outcomes.

3. When a predicted fault is confirmed during inspection or maintenance, the corresponding event is associated with the sensor data that preceded the fault.
4. The resulting information is supplied to the AI/ML model for updating or retraining the equipment-monitoring model.
5. This feedback mechanism enables the monitoring framework to progressively adapt to the individual operating characteristics of an industrial asset and improve its future monitoring and prediction capability.